

AI Automation, Regime Type, and Crossed Fragility

From Sketch to Global Games: Revising a Formal Model

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Motivation

The Question

How does AI automation affect regime stability?

The economic premise

Automation can follow different *trajectories* (Gans & Goldfarb 2026):

- ▶ **Rapid:** immediate displacement
- ▶ **Threshold:** complement first, substitute later

The political gap

Regime stability literature treats shocks by *magnitude*, not *shape*.

No model asks: *which trajectory is more dangerous for which regime?*

The Core Idea

The same AI technology can **stabilize** one regime while **destabilizing** the other.

Rapid displacement

Democracies compensate
Autocracies face organized revolt

Threshold automation

Democracies see no crisis until too late
Autocracies contain disoriented workers

This is the **crossed fragility** pattern.

Roadmap

Part I — The Original Paper

1. Model setup and v1 overview
2. What worked, what didn't

Part II — The Revision

3. Strategy and key solutions
4. Before/after comparison

Part III — The Welfare State

5. Can democracies protect themselves?
6. Social insurance as institutional insurance

Part IV — Verification & Lessons

7. Lean 4 formalization
8. What we learned

Part I: The Original Paper

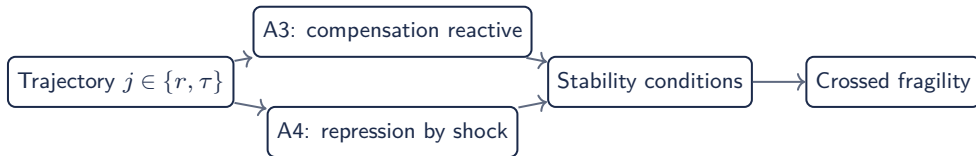
The Setup: What the Model Is About

A polity has **exposed workers** (E) facing AI automation over **two periods**. Regime is **democracy** (D) or **autocracy** (A). Workers choose: accept status quo (M) or protest (P).

Symbol	Meaning	Intuition
ω_t	Latent severity of automation	How bad the disruption really is
ℓ_t	Displacement (0 or L)	Did workers lose jobs?
φ_t	Democratic compensation	How much democracy offsets loss
κ_0, η_r	Repressive capacity $\times$ effectiveness	How well autocracy contains dissent
σ	Signal noise (by trajectory)	How predictable is the shock?
π^*	Equilibrium protest participation	Can workers coordinate?
β	Complementarity gain (≥ 0)	AI temporarily boosts income
w_E	Exposed worker's baseline wage	Income before automation
θ, W, k	Success share, total output, transition cost	Gains and costs of regime change
Δ	$\theta W - k - w_E$ (net revolt payoff)	Negative $\rightarrow$ no revolt without AI

Core asymmetry: Democracy stabilizes by making M *attractive* (compensation). Autocracy by making P *costly* (repression).

v1: Model Architecture



- ▶ Two periods, two trajectories, two regimes
- ▶ Five behavioral/structural assumptions (A1–A5)
- ▶ Binary displacement: $\ell_t \in \{0, L\}$
- ▶ Short, direct proofs

v1: The Assumptions

	Content	Plain language
A1	$\Delta < 0$	Without automation, no one revolts
A2	$L > \Delta $	The shock is big enough to trigger revolt
A3	$\varphi_t = \bar{\varphi}$ if displaced, else 0	Democracy only compensates during visible crises
A4	$\eta_r < 1, \eta_\tau = 1$	Repression degrades under rapid, holds under threshold
A5	Capacity built in $t=1$ persists	Institutions are slow to build

A3 and A4 are the **load-bearing** assumptions — and both are *directly imposed*, not derived from any deeper mechanism.

v1: Results at a Glance

	Content	Role
P1	Democratic fragility pattern	Core
P2	Autocratic fragility pattern	Core
P3	Crossed fragility	Core
P4	Welfare cost of autocratic stability	Core
P5–P6	Threshold of thresholds, interval width	Secondary
P7–P8	Welfare state as insurance	Secondary
P9	Endogenous fiscal fragility	Extension

v1: What Worked

Strengths

- ▶ Clear, testable result (crossed fragility)
- ▶ Parsimonious — 15 pages, clean proofs
- ▶ Good numerical example
- ▶ Welfare state extension is natural
- ▶ Five-type regime typology

The foundation was solid

The *intuition* was right: trajectory determines which regime is fragile.

The *formalization* needed work: too short a distance from assumptions to results.

Part I-B: Problems

Problem 1: Behavioral Assumptions Imposed

A3: compensation
is reactive

Why?

A4: repression depends
on shock type

Why?

- ▶ A3 *assumes* democracies only compensate when job losses are visible — but *why* do democracies behave this way? No mechanism given.
- ▶ A4 *assumes* repression works well against surprised workers but not organized ones — but *why*? Just postulated.
- ▶ Both describe exactly **the pattern we want to explain**, stated as premises
- ▶ A referee could say: “the crossed fragility is *assumed*, not proven”

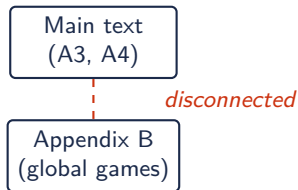
Problem 2: Binary Shock — No Microfoundation

In v1, the shock severity (ω , “how bad is automation?”) was binary — workers are either displaced or not:

$$\ell_t \in \{0, L\}$$

- ▶ No continuous *signal space* for workers to form beliefs about others
- ▶ Global games (Morris & Shin 2003) need continuous ω for unique equilibrium
- ▶ Appendix B *sketched* coordination but main proofs didn't *use* it

The gap:



Problems 3–4: Short Distance and Sparse Presentation

Short inferential distance



Results should *emerge* from deeper primitives, not be *restated* as assumptions.

Sparse presentation

v1 had:

- ▶ No figures in main text
- ▶ No mechanism diagram
- ▶ No coordination plot
- ▶ No income path visualization

Reader must build mental model entirely unaided.

Part II: The Revision

Revision Strategy

One primitive, everything derived.

Goal: Replace behavioral A3/A4 with a *single informational primitive* (σ) from which both compensation and repression emerge as equilibrium outcomes.

Tool: Global games (Morris & Shin 2003) moved from appendix to core.

Changes:

1. Shock severity ω : binary $\rightarrow$ continuous
2. Compensation & repression: assumed $\rightarrow$ derived from coordination equilibrium π^*
3. Assumptions unbundled (5 $\rightarrow$ 7)
4. Income paths: absorbing state + β (AI temporarily boosts income before displacing)
5. Four new figures
6. Formal Lean 4 verification

Solution 1: Continuous ω + Global Games

*How do we make compensation and repression **emerge** from a single mechanism?*

New information structure

Each worker observes a *private signal*:

$$s_{it} = \omega_t + \sigma_j \varepsilon_{it}$$

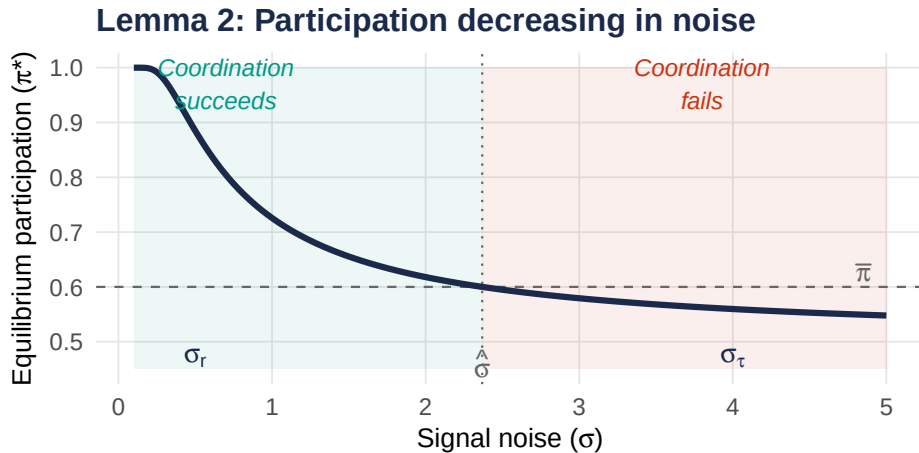
- ▶ ω_t : true severity (now continuous)
- ▶ σ_j : noise, depends on trajectory
- ▶ ε_{it} : individual perception error

If $\omega = 0.8$ and σ is low (rapid), worker sees $s_i \approx 0.8$. If σ is high (threshold), she might see 0.3 or 1.5 — she can't tell how bad things are.

Why this works

- ▶ Continuous $\omega \rightarrow$ Morris-Shin uniqueness theorem
- ▶ Uniform prior $\rightarrow$ Laplacian property $\rightarrow$ closed-form statics
- ▶ σ is the *single* trajectory-varying primitive
- ▶ Everything follows from $\pi^*(\sigma)$

Solution 1 (cont.): The Coordination Curve



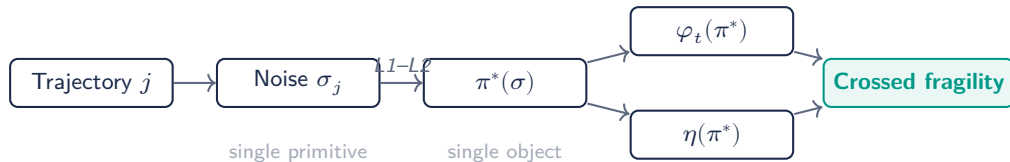
$\bar{\pi}$: minimum participation for collective action to succeed. $\hat{\sigma}$: critical noise at which coordination breaks down. Curve plotted at $\omega_0 = 1$ (realized shock severity when displacement occurs).

Solution 2: Assumptions Unbundled

v1			v2	
A3	Compensation reactive	→	A3	Information structure (M-S)
			A4	Majority-cost condition
			A5	Risk dominance
A4	η_r by shock type	→	A6	Trajectory separation
A5	Capacity is a stock	→	A7	Institutional lead time

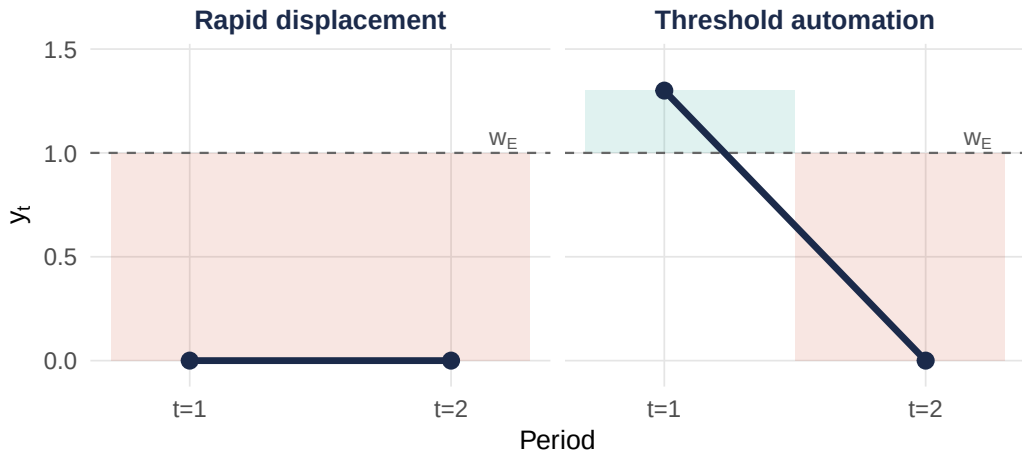
Key insight: The old behavioral A3 *bundled* three distinct conditions — the information structure, the voting game, and the risk dominance selection. Unbundling makes each testable and transparent.

Solution 3: The New Architecture



The **inferential chain** is now: trajectory $\rightarrow \sigma \rightarrow \pi^* \rightarrow$ institutional map $\rightarrow$ stability.
One primitive, one equilibrium, two outcomes.

Solution 4: Income Paths Redesigned



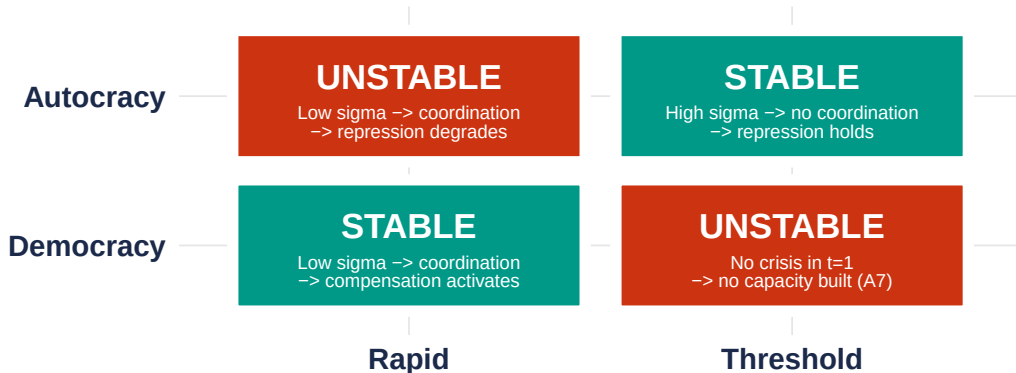
v1: $\ell_1 = L, \ell_2 = 0$ vs. $\ell_1 = 0, \ell_2 = L$. v2: rapid is an **absorbing state** (loss persists); threshold has a **complementarity gain** β (AI boosts income before replacing workers).

Solution 5: The Four Scenarios

Before I show you: which cell do you think is unstable? Democracy under rapid, or under threshold?

Crossed fragility (Proposition 3)

Moderate autocracy: $\bar{\kappa} \leq \kappa_0 < \bar{\kappa}/\eta_R$



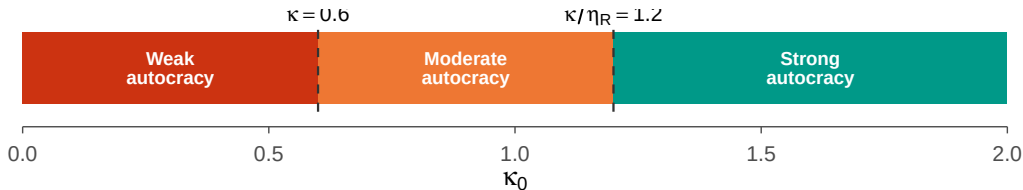
Part II-B: Before vs. After

Propositions: Before and After

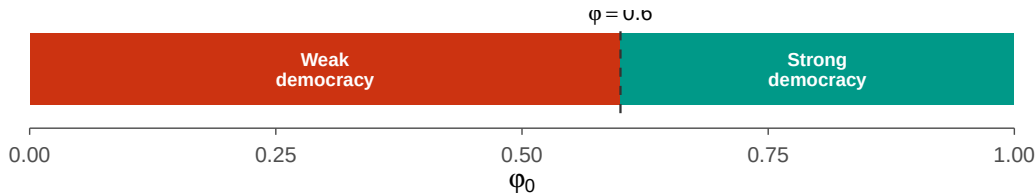
	Content	v1	v2
P1	Democratic fragility	Uses A3 directly	Uses L1–L2 + A7
P2	Autocratic fragility	Uses A4 directly	Uses L1–L2
P3	Crossed fragility	Combines A3+A4	Single π^* object
P4	Welfare cost	Same	Same
R1	Threshold of thresholds	Was P5	Now a Remark
R2	Interval width	Was P6	Now a Remark
P5	Welfare state insurance	Was P7	Rewritten with φ_0
P6	Functional equivalence	Was P8	Tighter proof
P7	Fiscal fragility	Was P9	Endogenous constraint

The Parametric Space

A. Autocracy stability



B. Democracy stability: $\bar{\varphi}$ =min. compensation to prevent revolt



The Complete Typology

Complete regime typology (Corollary 2)

Strong democracy	Stable	Stable	Crossed fragility (P3)
Weak democracy	Stable	Unstable	
Weak autocracy	Unstable	Unstable	
Moderate autocracy	Unstable	Stable	
Strong autocracy	Stable	Stable	
	Rapid	Threshold	

Part III: The Welfare State

Can Democracies Protect Themselves?

The new architecture reveals a natural question: if the coordination bottleneck is what makes democracies fragile under threshold automation, can they **bypass** it?

The problem

Weak democracies need a visible crisis to trigger compensation. Threshold shocks provide no visible crisis until too late (A7).

The solution

Build *standing* compensation (φ_0) *before* the shock — a welfare state that does not require crisis activation.

φ_0 is the democratic analogue of κ_0 (standing repression).

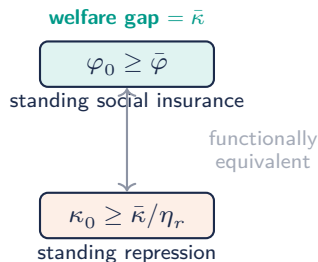
Insurance Against Delayed Shocks

P5: Democracy stable under threshold iff standing compensation $\varphi_0 \geq \bar{\varphi}$ (the compensatory threshold — how much compensation is needed to keep workers from revolting).

The welfare state **bypasses the coordination bottleneck**: it works even when workers can't organize.

P6: Both strong democracies and strong autocracies survive both trajectories. But the welfare gap is $\bar{\kappa} = L - |\Delta|$ (the repressive threshold):

$$u_E^D - u_E^A \geq \bar{\kappa} > 0$$



The Class Compromise Formalized

**Social insurance protects both the regime and its citizens.
Repression protects only the regime.**

Numerical example:

Strong democracy ($\varphi_0 = 0.6$):

$$u_E = w_E - 0.4$$

Strong autocracy ($\kappa_0 = 0.8$): $u_E = w_E - 1.0$

Gap: $0.6 = \bar{\kappa}$

Przeworski & Wallerstein (1982):

Workers consent to profit in exchange for protection.

Our formalization: When institutionalized, this compromise insures the *regime itself* against technological disruption.

Part IV: Verification & Lessons

Lean 4 Formal Verification

What we verified:

- ▶ P1–P7: all propositions
- ▶ R1–R2: remarks
- ▶ Coordination conditions
- ▶ Single-crossing, dominance regions

17/17 verified. Including L1–L2
(SupermodularGames lib).

Why it matters:

- ▶ Every algebraic step machine-checked
- ▶ Caught two sign errors in drafts
- ▶ Forces precision in assumptions
- ▶ Builds reviewer confidence

```
cd formal_proofs && lake build
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Comparative Statics Summary

Parameter	Effect	Interpretation
$L \uparrow$	$\bar{\varphi} \uparrow, \bar{\kappa} \uparrow$	Larger shocks destabilize both
$\eta_r \downarrow$	Crossed interval widens	More autocracies exhibit crossed pattern
$\theta \uparrow$	$\Delta \uparrow$, thresholds rise	Regime change more attractive
$k \uparrow$	$\Delta \downarrow$, thresholds fall	Regime change costlier
$\kappa_0 \uparrow$	Autocracy more stable	Weak $\rightarrow$ moderate $\rightarrow$ strong
$\varphi_0 \uparrow$	Democracy more stable (threshold)	Welfare state as insurance

Lessons from the Revision

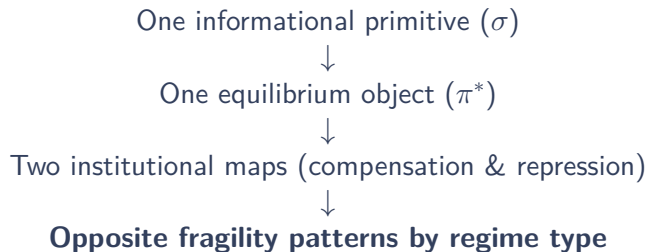
On formal modeling

1. **Derive, don't assume.** Make behavioral patterns emerge from deeper primitives.
2. **Unbundle assumptions.** Each should do one job.
3. **Figures are arguments.** Mechanism diagrams are proof strategy, not decoration.

On the process

4. **Start with the example.** The numerical example was the anchor throughout.
5. **Verify formally.** Lean caught errors re-reading missed.
6. **The revision is the paper.** v1 found the result; v2 *proved* it.

Crossed fragility is not assumed — it emerges.



The welfare state is the democratic substitute for repression.